📘 Chapter 3: Water Resources – Class 10 Geography Notes (CBSE)

🌎 1. Water as a Renewable Resource

- About three-fourths of the Earth's surface is covered with water, but only a small part is fresh water suitable for human use.
- Freshwater mainly comes from surface runoff and groundwater, which are constantly renewed by the **hydrological cycle**.
- The hydrological cycle involves evaporation, condensation, precipitation, and infiltration, ensuring that water is a **renewable natural resource**.

💧 2. Water Scarcity and Need for Conservation

📌 What is Water Scarcity?

Water scarcity is the lack of sufficient available water resources to meet the demands of water usage within a region.

🚫 Causes of Water Scarcity

1. **Over-exploitation and excessive use:** Rapid population growth increases water demand for domestic use, food production, and industry.
2. **Agricultural demands:** Irrigated agriculture is the largest consumer of water. Over-extraction of groundwater leads to declining water levels.
3. **Industrialisation:** Growing industries consume vast amounts of water and energy, often leading to pollution.
4. **Urbanisation:** Expanding cities with dense populations put enormous pressure on water resources.
5. **Water pollution:** Industrial and domestic wastes, pesticides, and fertilisers pollute freshwater sources.
6. **Unequal access:** Social and economic inequalities often result in unequal distribution of water.
7. **Poor quality of water:** Even if water is available, it may be unsuitable for use if contaminated.

⚠️ Consequences of Water Scarcity

- Health problems due to polluted water.
- Threat to food security due to water shortage in agriculture.
- Loss of livelihoods and reduction in economic activities.
- Ecological imbalance and degradation of ecosystems.

🏛️ Government Initiatives

- **Jal Jeevan Mission (JJM):** Ensures every rural household receives 55 litres of safe drinking water per person per day.
- **Atal Bhujal Yojana (Atal Jal):** Promotes community-driven water conservation and smart water management.

🏗️ 3. Multi-purpose River Projects and Integrated Water Resource Management

- Since ancient times, people in India have built hydraulic structures like dams, reservoirs, canals, and embankments.
- Today, large dams serve many purposes and are known as **multi-purpose river projects**.

🌊 Uses of Multi-purpose Projects

- Irrigation
- Hydroelectric power generation
- Domestic and industrial water supply
- Flood control
- Recreation
- Inland navigation
- Fish breeding

📌 Examples:

- **Bhakra-Nangal Project:** Sutlej-Beas basin – used for hydropower and irrigation.
- **Hirakud Dam:** Mahanadi basin – used for flood control and water conservation.

✅ Advantages of Multi-purpose Projects

- Supply water to areas suffering from scarcity.
- Help control floods by regulating water flow.
- Generate hydroelectric power.
- Support agriculture and industrial development.
- Provide facilities for navigation and fish farming.

❌ Disadvantages of Large Dams

- Disturb the natural flow and sediment transport of rivers.
- Cause excessive sedimentation and rocky stream beds.
- Block fish migration and affect aquatic life.
- Submerge vegetation and soil, causing decomposition and pollution.
- Sometimes trigger floods due to siltation in reservoirs.
- Cause soil erosion and loss of natural fertiliser from silt.
- Lead to water-borne diseases and earthquakes.
- Change cropping patterns, leading to salinisation of soil.

🌾 Government Initiative

- **Pradhan Mantri Krishi Sinchayee Yojana (PMKSY):**
  - "**Har Khet Ko Pani**" – irrigation for every farm.
  - "**Per Drop More Crop**" – improve water use efficiency.

- Promote sustainable water conservation practices.

🏞️ **Example: Sardar Sarovar Dam (Narmada River)**

- Covers Gujarat, Madhya Pradesh, Maharashtra, and Rajasthan.
- Provides irrigation to 18.45 lakh hectares in Gujarat and other states.
- Helps drought-prone areas become drought-proof.

⚖️ **Inter-State Water Disputes**

- **Krishna-Godavari Dispute:** Maharashtra's diversion of water at Koyna affects water availability in Karnataka and Andhra Pradesh.

🌧️ **4. Rainwater Harvesting**

Rainwater harvesting is the collection and storage of rainwater for future use. It is an eco-friendly and sustainable way of managing water resources.

🏞️ **Traditional Methods of Rainwater Harvesting**

- **Guls or Kuls:** Diversion channels in Western Himalayas.
- **Rooftop Rainwater Harvesting:** Used in Rajasthan for storing drinking water.
- **Inundation Channels:** Used in Bengal's floodplains for irrigation.
- **Khadins and Johads:** Rain-fed storage structures in Rajasthan.

🏠 **Rooftop Rainwater Harvesting (Tankas)**

- Common in Bikaner, Phalodi, and Barmer.
- Underground tanks connected to sloping roofs collect rainwater.
- The first rainfall is allowed to clean roofs and pipes.
- Later rainfall is stored for drinking water.
- **Palar pani** (rainwater) is considered the purest natural water.
- Underground rooms near tankas stay cool in summer.

📌 **Example: Gendathur Village, Karnataka**

- About 200 houses have rooftop systems.
- Each house collects ~50,000 litres annually.
- Total rainwater harvested = ~1,00,000 litres per year.

🏛️ **Modern Rainwater Harvesting Practices**

- **Tamil Nadu:** First state to make rooftop harvesting legally mandatory.
- **Shillong, Meghalaya:** Rooftop harvesting meets 15–25% of household water needs despite heavy rainfall nearby.

🎋 Bamboo Drip Irrigation (Meghalaya)

- A 200-year-old method to transport water using bamboo pipes.
- Water is diverted by gravity from hilltop springs to lower areas.
- Flow reduces from 18–20 litres to 20–80 drops per minute near plants.
- Simple, cost-effective, and sustainable.

📚 Important Questions & Answers

📝 30 Words

Q1. How is water a renewable resource?
Water is renewed through the **hydrological cycle** involving evaporation, condensation, precipitation, and infiltration, ensuring continuous availability of surface and groundwater.

Q2. What is water scarcity and what are its causes?
Water scarcity is the lack of sufficient water to meet human needs. Causes include overuse, over-extraction, pollution, industrialisation, population growth, and unequal access.

📝 120 Words

Q1. Describe how rainwater harvesting is carried out in semi-arid regions of Rajasthan.
In Rajasthan, houses have **underground tankas** connected to sloping roofs to collect rainwater. The first rain is used to clean the roofs and pipes, and subsequent rain is stored. This collected rainwater, called **palar pani**, is considered pure and is used for drinking when other sources dry up. Tankas are part of the house design and help cool adjoining rooms. Rainwater harvesting ensures water availability during summers and droughts. Even with the Indira Gandhi Canal, some households still use this system due to their preference for rainwater's taste.

Q2. How are traditional rainwater harvesting methods adapted today?
Traditional techniques like rooftop rainwater harvesting and bamboo drip irrigation are now adapted with modern materials like PVC pipes and filtration systems. Rooftop harvesting is compulsory in Tamil Nadu and widely used in Shillong. Villages like Gendathur use these systems to collect large quantities of water annually. Bamboo drip irrigation is still used in Meghalaya to transport spring water sustainably. These adaptations help conserve water, reduce dependence on external sources, and ensure sustainable water use in both rural and urban areas.

📊 Glossary

- **Water scarcity:** Shortage of water resources to meet human needs.
- **Hydrological cycle:** Continuous movement of water on, above, and below the Earth's surface.
- **Multi-purpose river project:** A dam project serving multiple purposes like irrigation, power generation, and flood control.
- **Rainwater harvesting:** Collecting and storing rainwater for future use.
- **Tanka:** Underground water storage tank used in Rajasthan.
- **Palar pani:** Term for pure rainwater in Rajasthan.
- **Bamboo drip irrigation:** A traditional water transport system in Meghalaya using bamboo pipes.

🗺️ Map Work: Major Rivers and Dams of India

You must know the location of the following for map-based questions:

- Bhakra-Nangal Dam – Sutlej River (Himachal Pradesh)
- Hirakud Dam – Mahanadi River (Odisha)
- Sardar Sarovar Dam – Narmada River (Gujarat)
- Tehri Dam – Bhagirathi River (Uttarakhand)
- Nagarjuna Sagar Dam – Krishna River (Telangana)